

LAB 1 — USE-CASE FLOW

Smart Lab Equipment & Slot Reservation Portal

Use Case ID: UC-03

Use Case Name: Reserve Equipment & Slot

Primary Actor: Student

Preconditions

1. Student is authenticated in the system.
2. Lab equipment is available in the portal.
3. The selected equipment is eligible for reservation.
4. The requested slot is available.
5. The requested reservation satisfies the system's booking rules.

Postconditions

1. The reservation is stored successfully.
2. The selected equipment and slot are reserved for the student.
3. A reservation confirmation/token is generated.
4. The reserved slot is unavailable for another reservation.

Main Success Scenario

1. Student logs into the Smart Lab Equipment & Slot Reservation Portal.
2. Student selects Reserve Equipment & Slot.
3. System displays available lab equipment.
4. Student selects the required equipment.
5. System displays available reservation slots.
6. Student selects an available slot.
7. System validates the selected equipment and slot.
8. System creates the reservation.
9. System generates a reservation confirmation/token.
10. System displays the reservation details to the student.
11. Use case ends successfully.

Alternate Flow — Slot Unavailable

6a. Selected slot is no longer available

1. System detects that the selected slot is unavailable.
2. System rejects the reservation request.
3. System displays an appropriate message.
4. System displays the currently available slots.
5. Student selects another available slot.
6. System continues from Step 7 of the Main Success Scenario.